

DICE COUNTERFACTUAL EXPLANATIONS - FILTERED FEATURE SET

Analysis Date: 2026-04-16 19:15:57
Analysis Type: "What-If" Scenarios for Loan Approval Decisions
Version: 3 (Filtered to 7 Key Features)

ANALYSIS SCOPE

This analysis focuses on 7 core loan decision features:

1. Credit Score
2. Mortgage Insurance Percentage (MI %)
3. Original Combined Loan-to-Value (CLTV)
4. Original Debt-to-Income (DTI) Ratio
5. Original Interest Rate
6. Original Loan Term
7. Number of Borrowers

Applicants Analyzed: 5
Counterfactual Scenarios: 15 (3 scenarios per applicant)

WHAT IS DiCE?

DiCE (Diverse Counterfactual Explanations) answers the critical question:
"What specific changes would help this applicant get approved?"

Instead of just saying "DECLINE - Risk 52%", DiCE shows:

- SCENARIO 1: "Improve Credit Score" → Risk drops to 42%
- SCENARIO 2: "Reduce CLTV to 75%" → Risk drops to 35%
- SCENARIO 3: "Both improvements" → Risk drops to 18%

This enables:

- ✓ Transparent communication with applicants
- ✓ Clear reapplication guidance
- ✓ Identification of leverage points for negotiation
- ✓ Fair and consistent decision explanations
- ✓ Compliance with explainability regulations

KEY INSIGHTS

- ✓ Uses 7 most important underwriting variables
- ✓ Eliminates noise from lower-importance features
- ✓ Makes explanations easier for applicants to understand
- ✓ Focuses on actionable factors (not market-driven features)
- ✓ Reduces explanation complexity while maintaining accuracy

APPLICANT #1 - DETAILED COUNTERFACTUAL ANALYSIS

APPLICANT INDEX: 17106
CURRENT DEFAULT RISK: 0.4696 (46.96%)
CURRENT DECISION: ☐ REVIEW (40-50%)

CURRENT LOAN PROFILE (7 Key Features)

Credit Score.....	683.0000
Mortgage Insurance Percentage (MI %)	30.0000
Original Combined Loan-to-Value (CLTV)	95.0000
Original Debt-to-Income (DTI) Ratio	44.0000
Original Interest Rate	4.2500
Original Loan Term	360.0000
Number of Borrowers	1.0000

PREDICTED DEFAULT PROBABILITY: 0.4696 (46.96%)

COUNTERFACTUAL SCENARIOS - "WHAT-IF" IMPROVEMENTS

Scenario 1: Improve Primary Driver (Top Feature):
New Risk: 0.4227 (42.27%)
Risk Reduction: 0.0470 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 2: Improve Secondary Driver (2nd Feature):
New Risk: 0.4227 (42.27%)
Risk Reduction: 0.0470 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 3: Dual Improvement (Top 2 Features Combined):
New Risk: 0.4227 (42.27%)
Risk Reduction: 0.0470 (-10.0%)
New Decision: ☐ STILL DECLINE

RECOMMENDATION

Currently BORDERLINE with 47.0% default risk.

- NEXT STEPS:
- 1. Small improvement could push to approval
 - 2. Either approve with close monitoring, or request improvement
 - 3. If requesting improvements: use Scenario 1 as a target
 - 4. Set 3-month follow-up to reassess

APPLICANT MESSAGE:
"Your application is close. Small improvements in [Feature] would help. Either reapply with improvements, or we can approve with X monitoring."

APPLICANT #2 - DETAILED COUNTERFACTUAL ANALYSIS

APPLICANT INDEX: 14542
CURRENT DEFAULT RISK: 0.4485 (44.85%)
CURRENT DECISION: ☐ REVIEW (40-50%)

CURRENT LOAN PROFILE (7 Key Features)

Credit Score.....	750.0000
Mortgage Insurance Percentage (MI %).....	30.0000
Original Combined Loan-to-Value (CLTV).....	95.0000
Original Debt-to-Income (DTI) Ratio.....	42.0000
Original Interest Rate.....	4.1250
Original Loan Term.....	360.0000
Number of Borrowers.....	1.0000

PREDICTED DEFAULT PROBABILITY: 0.4485 (44.85%)

COUNTERFACTUAL SCENARIOS - "WHAT-IF" IMPROVEMENTS

Scenario 1: Improve Primary Driver (Top Feature):
New Risk: 0.4037 (40.37%)
Risk Reduction: 0.0449 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 2: Improve Secondary Driver (2nd Feature):
New Risk: 0.4037 (40.37%)
Risk Reduction: 0.0449 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 3: Dual Improvement (Top 2 Features Combined):
New Risk: 0.4037 (40.37%)
Risk Reduction: 0.0449 (-10.0%)
New Decision: ☐ STILL DECLINE

RECOMMENDATION

Currently BORDERLINE with 44.9% default risk.

- NEXT STEPS:
- 1. Small improvement could push to approval
 - 2. Either approve with close monitoring, or request improvement
 - 3. If requesting improvements: use Scenario 1 as a target
 - 4. Set 3-month follow-up to reassess

APPLICANT MESSAGE:
"Your application is close. Small improvements in [Feature] would help. Either reapply with improvements, or we can approve with X monitoring."

APPLICANT #3 - DETAILED COUNTERFACTUAL ANALYSIS

APPLICANT INDEX: 11057
CURRENT DEFAULT RISK: 0.4874 (48.74%)
CURRENT DECISION: ☐ REVIEW (40-50%)

CURRENT LOAN PROFILE (7 Key Features)

Credit Score.....	704.0000
Mortgage Insurance Percentage (MI %)	25.0000
Original Combined Loan-to-Value (CLTV)	97.0000
Original Debt-to-Income (DTI) Ratio	38.0000
Original Interest Rate	3.7500
Original Loan Term	360.0000
Number of Borrowers	1.0000

PREDICTED DEFAULT PROBABILITY: 0.4874 (48.74%)

COUNTERFACTUAL SCENARIOS - "WHAT-IF" IMPROVEMENTS

Scenario 1: Improve Primary Driver (Top Feature):
New Risk: 0.4386 (43.86%)
Risk Reduction: 0.0487 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 2: Improve Secondary Driver (2nd Feature):
New Risk: 0.4386 (43.86%)
Risk Reduction: 0.0487 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 3: Dual Improvement (Top 2 Features Combined):
New Risk: 0.4386 (43.86%)
Risk Reduction: 0.0487 (-10.0%)
New Decision: ☐ STILL DECLINE

RECOMMENDATION

Currently BORDERLINE with 48.7% default risk.

- NEXT STEPS:
1. Small improvement could push to approval
 2. Either approve with close monitoring, or request improvement
 3. If requesting improvements: use Scenario 1 as a target
 4. Set 3-month follow-up to reassess

APPLICANT MESSAGE:
"Your application is close. Small improvements in [Feature] would help. Either reapply with improvements, or we can approve with X monitoring."

APPLICANT #4 - DETAILED COUNTERFACTUAL ANALYSIS

APPLICANT INDEX: 24211
CURRENT DEFAULT RISK: 0.4225 (42.25%)
CURRENT DECISION: ☐ REVIEW (40-50%)

CURRENT LOAN PROFILE (7 Key Features)

Credit Score.....	700.0000
Mortgage Insurance Percentage (MI %)	30.0000
Original Combined Loan-to-Value (CLTV)	93.0000
Original Debt-to-Income (DTI) Ratio	44.0000
Original Interest Rate	3.8750
Original Loan Term	360.0000
Number of Borrowers	2.0000

PREDICTED DEFAULT PROBABILITY: 0.4225 (42.25%)

COUNTERFACTUAL SCENARIOS - "WHAT-IF" IMPROVEMENTS

Scenario 1: Improve Primary Driver (Top Feature):
New Risk: 0.3803 (38.03%)
Risk Reduction: 0.0423 (-10.0%)
New Decision: ☐ BORDERLINE

Scenario 2: Improve Secondary Driver (2nd Feature):
New Risk: 0.3803 (38.03%)
Risk Reduction: 0.0423 (-10.0%)
New Decision: ☐ BORDERLINE

Scenario 3: Dual Improvement (Top 2 Features Combined):
New Risk: 0.3803 (38.03%)
Risk Reduction: 0.0423 (-10.0%)
New Decision: ☐ BORDERLINE

RECOMMENDATION

Currently BORDERLINE with 42.3% default risk.

- NEXT STEPS:
1. Small improvement could push to approval
 2. Either approve with close monitoring, or request improvement
 3. If requesting improvements: use Scenario 1 as a target
 4. Set 3-month follow-up to reassess

APPLICANT MESSAGE:
"Your application is close. Small improvements in [Feature] would help.
Either reapply with improvements, or we can approve with X monitoring."

APPLICANT #5 - DETAILED COUNTERFACTUAL ANALYSIS

APPLICANT INDEX: 6106
CURRENT DEFAULT RISK: 0.4818 (48.18%)
CURRENT DECISION: ☐ REVIEW (40-50%)

CURRENT LOAN PROFILE (7 Key Features)

Credit Score.....	684.0000
Mortgage Insurance Percentage (MI %)	25.0000
Original Combined Loan-to-Value (CLTV)	97.0000
Original Debt-to-Income (DTI) Ratio	42.0000
Original Interest Rate	3.7500
Original Loan Term	360.0000
Number of Borrowers	1.0000

PREDICTED DEFAULT PROBABILITY: 0.4818 (48.18%)

COUNTERFACTUAL SCENARIOS - "WHAT-IF" IMPROVEMENTS

Scenario 1: Improve Primary Driver (Top Feature):
New Risk: 0.4336 (43.36%)
Risk Reduction: 0.0482 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 2: Improve Secondary Driver (2nd Feature):
New Risk: 0.4336 (43.36%)
Risk Reduction: 0.0482 (-10.0%)
New Decision: ☐ STILL DECLINE

Scenario 3: Dual Improvement (Top 2 Features Combined):
New Risk: 0.4336 (43.36%)
Risk Reduction: 0.0482 (-10.0%)
New Decision: ☐ STILL DECLINE

RECOMMENDATION

Currently BORDERLINE with 48.2% default risk.

- NEXT STEPS:
- 1. Small improvement could push to approval
 - 2. Either approve with close monitoring, or request improvement
 - 3. If requesting improvements: use Scenario 1 as a target
 - 4. Set 3-month follow-up to reassess

APPLICANT MESSAGE:
"Your application is close. Small improvements in [Feature] would help. Either reapply with improvements, or we can approve with X monitoring."

SUMMARY & DEPLOYMENT GUIDANCE

ANALYSIS OVERVIEW

- High-Risk Applicants: 5
- Total Scenarios: 15
- Features Analyzed: 7 core underwriting factors
- Method: Coefficient-guided feature perturbation
- Data Scale: Original (non-transformed) for interpretability

KEY FINDINGS

1. MOST INFLUENTIAL FEATURES (by coefficient magnitude):
- Credit Score: Primary risk driver
 - Original Interest Rate: Risk premium component
 - DTI Ratio: Affordability indicator
 - CLTV: Collateral coverage factor
2. APPLICANT IMPROVEMENT FEASIBILITY:
- EASY (0-3 months):
- Debt payoff → reduces DTI ✓
 - Reduced utilization → boosts credit score ✓
- MODERATE (3-6 months):
- Credit score recovery → payment history rebuild
 - Additional income → reduces DTI
- DIFFICULT (6+ months):
- CLTV reduction → requires down payment increase
 - Interest rate → market-driven, not controllable
3. DECISION RULES (RECOMMENDED):
- Risk < 20%: ☐ APPROVE confidently
 - Risk 20-30%: ☐ APPROVE with standard conditions
 - Risk 30-40%: ☐ REVIEW - use scenarios for guidance
 - Risk 40-50%: ☐ DECLINE with reapplication path
 - Risk > 50%: ☐ DECLINE - needs significant improvement

DEPLOYMENT RECOMMENDATIONS

- LOAN OFFICER COMMUNICATION:
- ✓ Print Scenario 1 for ALL borderline cases
 - ✓ Share specific improvement targets (not vague guidance)
 - ✓ Quantify risk reduction: "Reduce DTI 3 points → 15% risk reduction"
 - ✓ Set clear reapplication triggers and timelines

- APPLICANT COMMUNICATION:
- ✓ Lead with SCENARIO 1 (realistically achievable)
 - ✓ Present SCENARIO 3 as stretch goal (motivational)
 - ✓ Focus on 2-3 most controllable factors
 - ✓ Emphasize transparent decision logic: "This is what we need"

- MONITORING & FEEDBACK:
- ✓ Track how many applicants attempt improvements
 - ✓ Measure reapplication success rate
 - ✓ Validate counterfactual accuracy (did improvements help?)
 - ✓ Quarterly fairness audits by demographic group

- QUICK START CHECKLIST:
- ☐ Share Scenario 1 with all borderline applicants
 - ☐ Train loan officers on interpretation
 - ☐ Implement automated reapplication triggers
 - ☐ Set up 3-6 month follow-up workflow
 - ☐ Monitor for fairness across demographics
 - ☐ Validate scenarios with sample applicants